DESIGN AND SIMULATION OF A SIMPLIFIED PCI EXPRESS GEN6 DIGITAL MODEL

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2026

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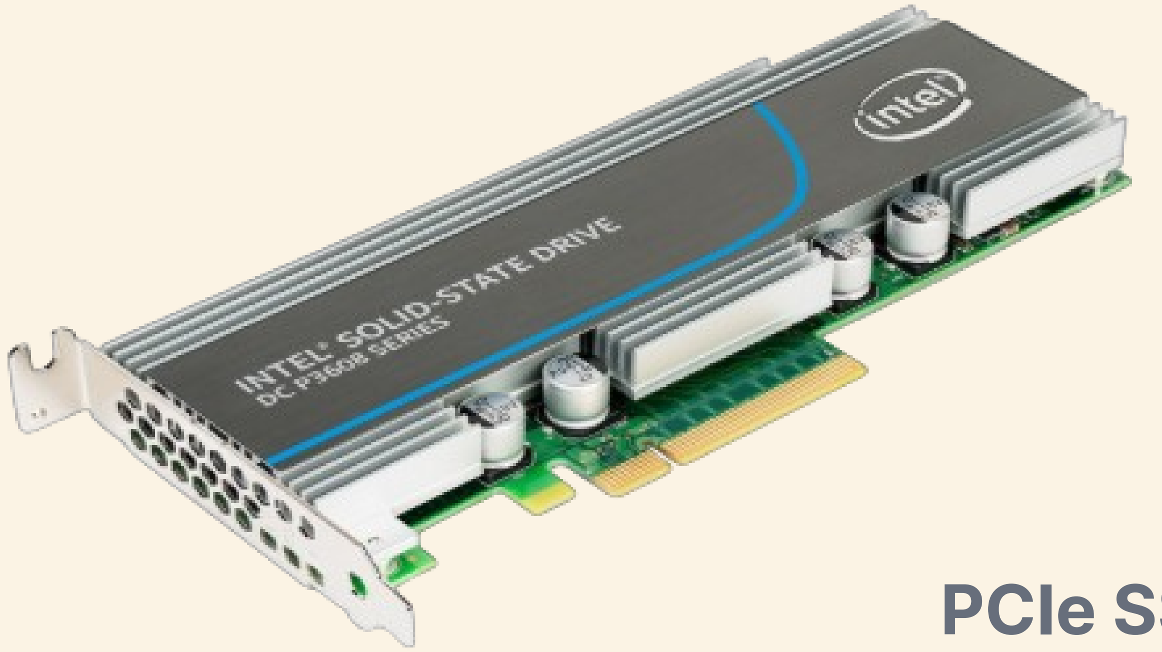
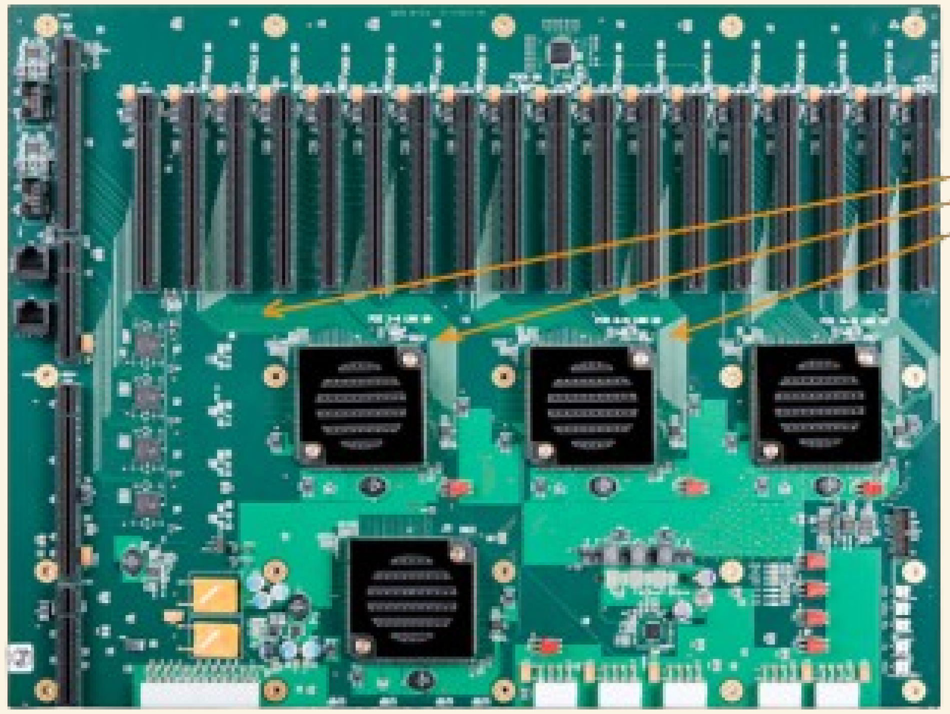
BACKGROUND

What is PCI Express (PCIe)?

- HIGH-SPEED INTERFACE CONNECTING DEVICES (GPU, SSD)
- USES LANES (X1–X16) FOR HIGHER SPEED
- LOW LATENCY FOR FAST COMMUNICATION
- MANAGED BY PCI-SIG



PCIe SSD



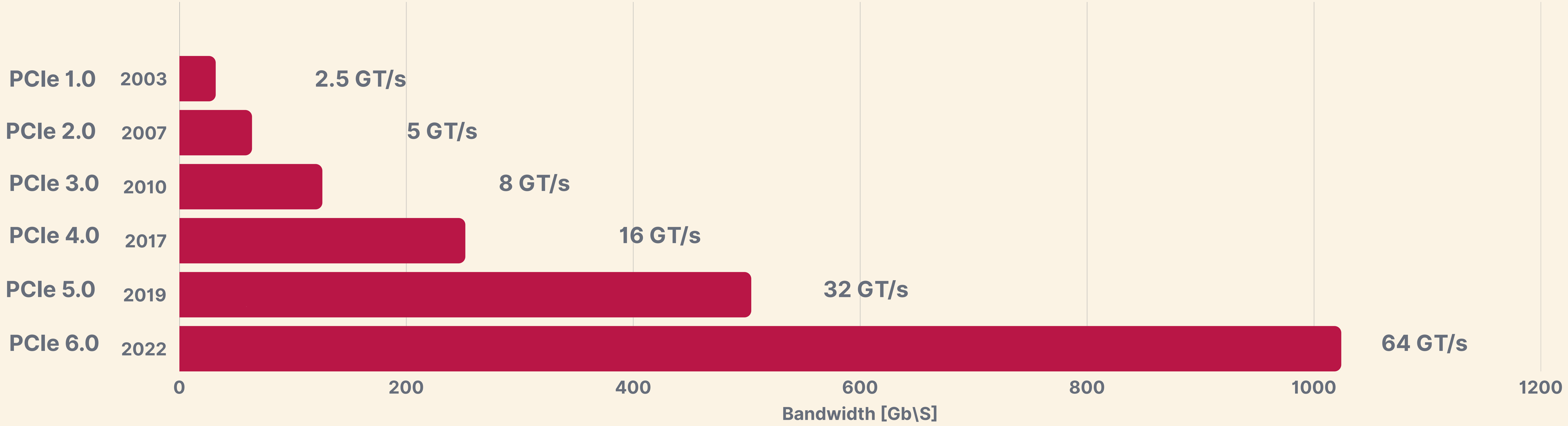
Motherboard PCIe Slots

MOTIVATION

Why PCIe 6.0

- **AI/ML EXPLOSION**
  GPT-scale models require >1 TB/s memory bandwidth. PCIe 6.0 x16 delivers 128 GB/s — 2× Gen5, enabling GPU-to-CPU latencies previously impossible.
- **DATA CENTER DEMAND**
  Hyperscalers (AWS, Google, Microsoft) deploy millions of PCIe endpoints. Gen6 doubles throughput at same power envelope via PAM4 + FEC instead of higher voltage swings.
- **CXL 3.0 FOUNDATION**
  Compute Express Link 3.0 is built on PCIe 6.0 electrical. Implementing PCIe 6.0 is foundational for next-gen memory-semantic fabrics.
- **ACADEMIC GAP**
  No open-source, fully-layered PCIe 6.0 RTL reference exists. This project fills that gap with a verifiable, documented implementation.

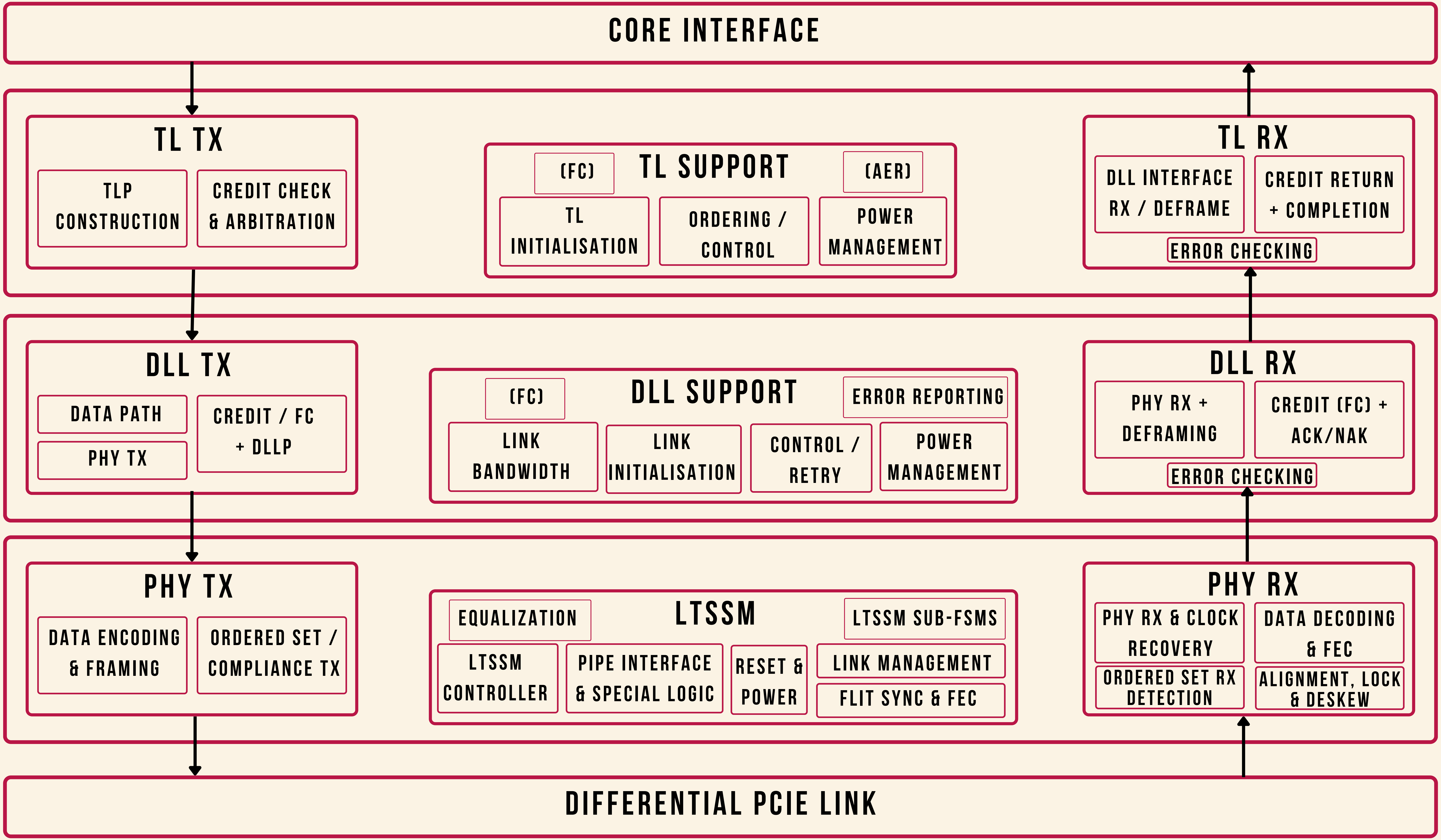
PCIE EVOLUTION



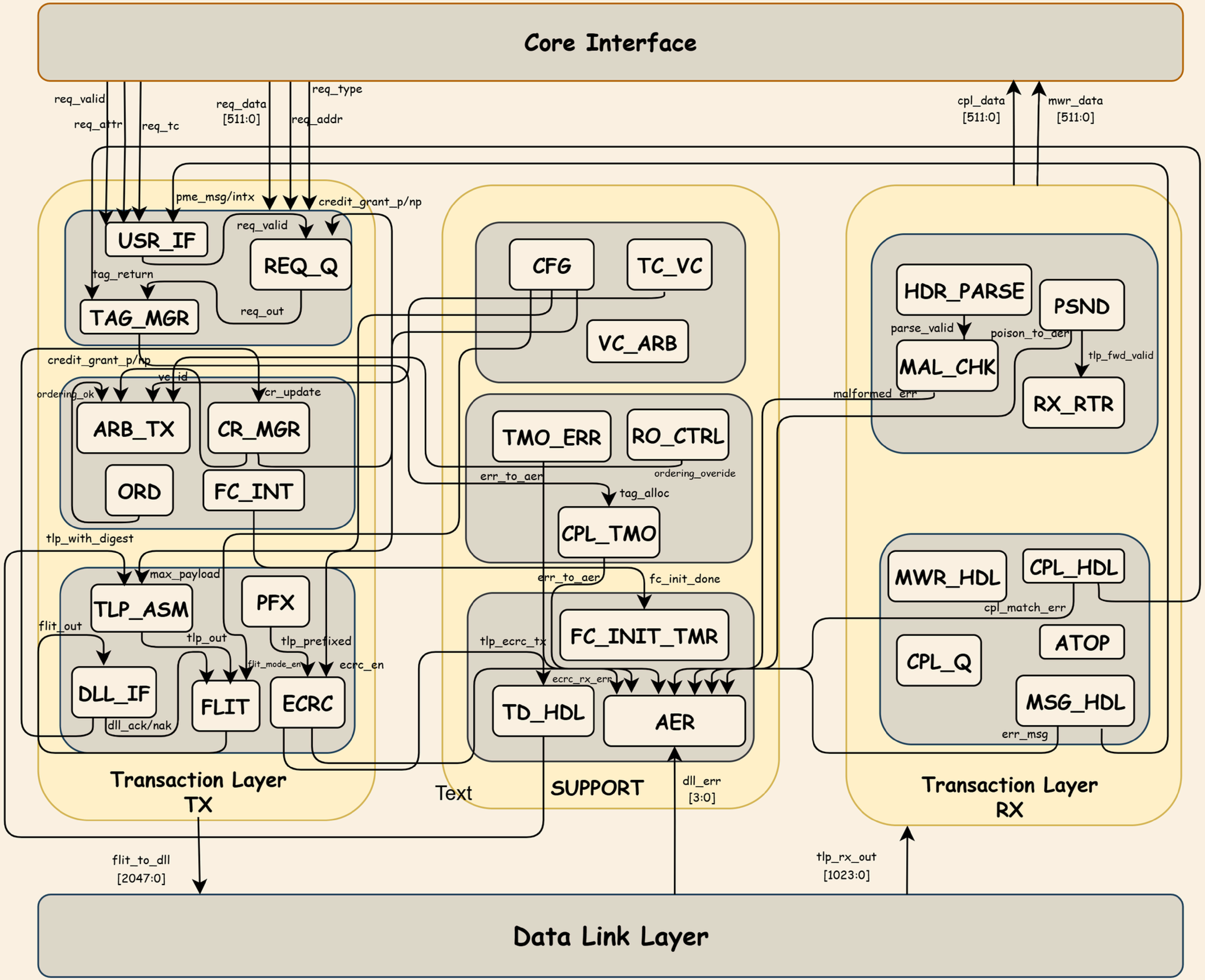
PCIE 6.0 KEY FEATURES

- 64 GT/S RAW DATA RATE
- PAM4 SIGNALING
- FLIT-BASED FRAMING
- MANDATORY FEC
- LOSSLESS FLOW CONTROL
- L0P LOW-POWER STATE
- CRC ENHANCEMENT
- BACKWARD COMPATIBILITY

SYSTEM ARCHITECTURE OVERVIEW

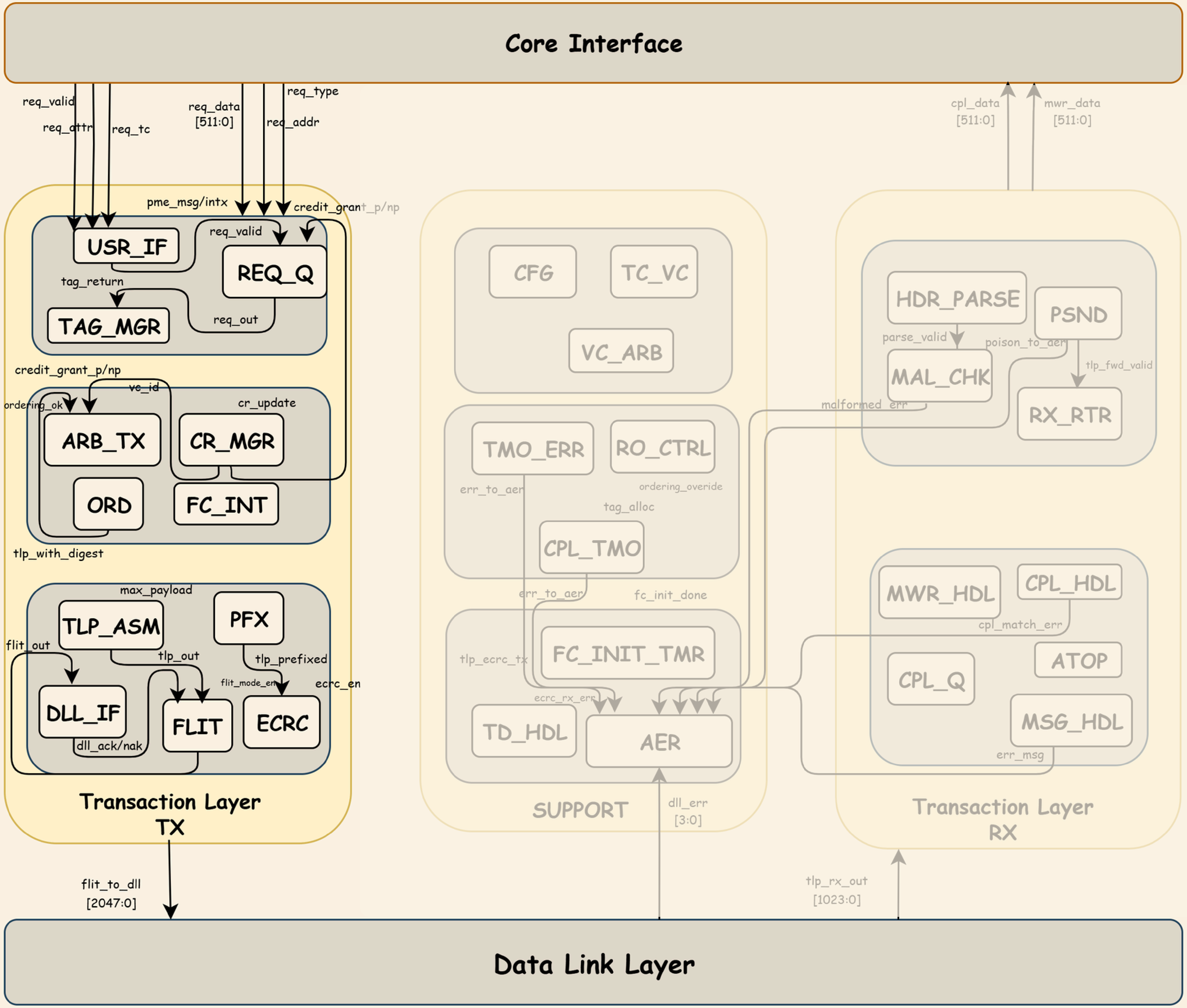

CORE INTERFACE
TL TX
TLP CONSTRUCTION
CREDIT CHECK & ARBITRATION
(FC)
TL SUPPORT
(AER)
TL INITIALISATION
ORDERING / CONTROL
POWER MANAGEMENT
TL RX
DLL INTERFACE RX / DEFRAME
CREDIT RETURN + COMPLETION
ERROR CHECKING
DLL TX
DATA PATH
PHY TX
CREDIT / FC + DLLP
(FC)
DLL SUPPORT
ERROR REPORTING
LINK BANDWIDTH
LINK INITIALISATION
CONTROL / RETRY
POWER MANAGEMENT
DLL RX
PHY RX + DEFRAMING
CREDIT (FC) + ACK/NAK
ERROR CHECKING
PHY TX
DATA ENCODING & FRAMING
ORDERED SET / COMPLIANCE TX
EQUALIZATION
LTSSM
LTSSM SUB-FSMS
LTSSM CONTROLLER
PIPE INTERFACE & SPECIAL LOGIC
RESET & POWER
LINK MANAGEMENT
FLIT SYNC & FEC
PHY RX
PHY RX & CLOCK RECOVERY
DATA DECODING & FEC
ORDERED SET RX DETECTION
ALIGNMENT, LOCK & DESKEW
DIFFERENTIAL PCIE LINK

TRANSACTION LAYER CORE ARCHITECTURE



TL TX

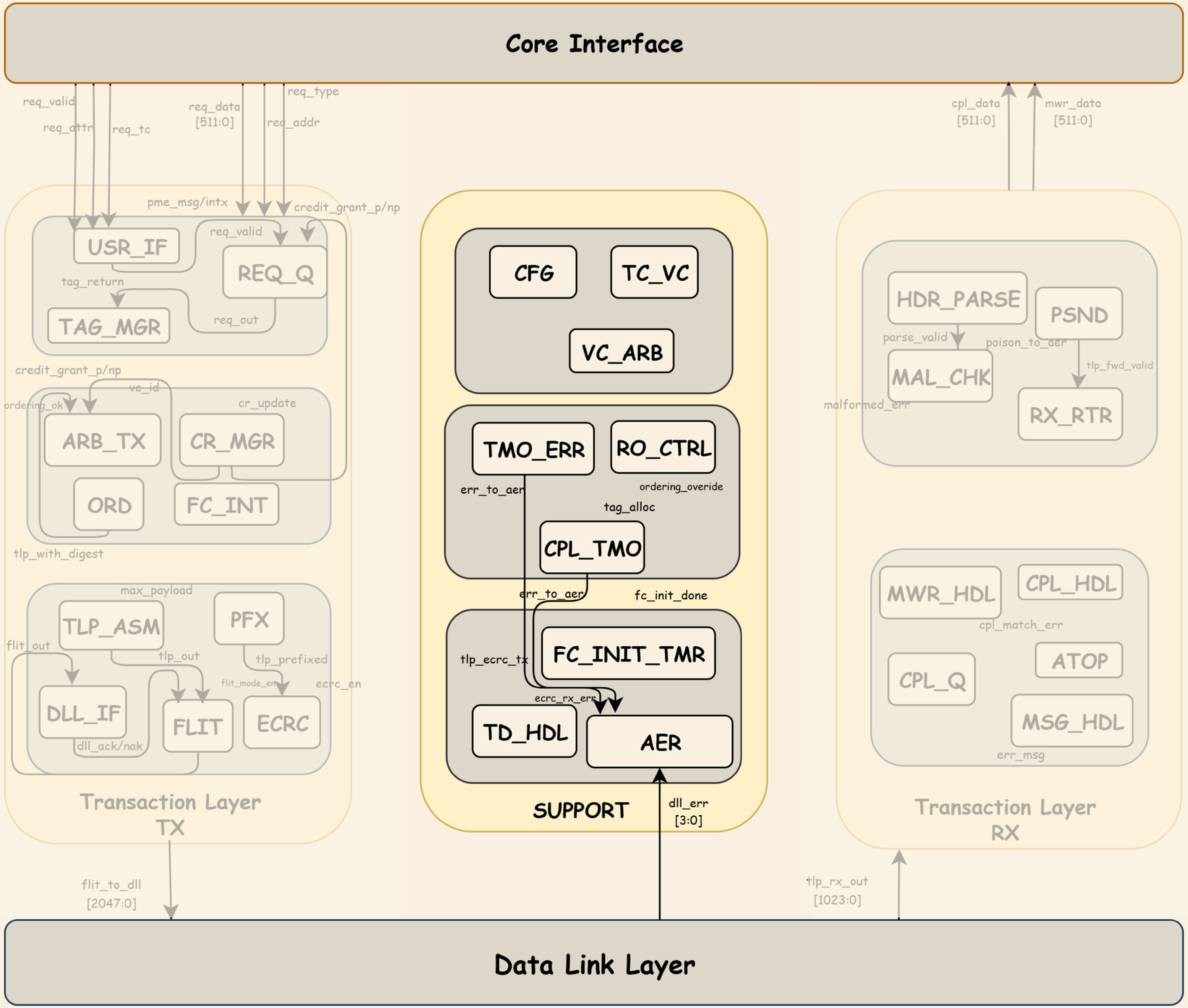
- Request receiving from processor.
- Tag and Credit managments .
- Assembling and arbitration .
- FLIT Preparation (TLPs).
- Error Detection Before Transmission.
- DLL INTERFACE



SUPPORT

FLOW CONTROL ENGINE

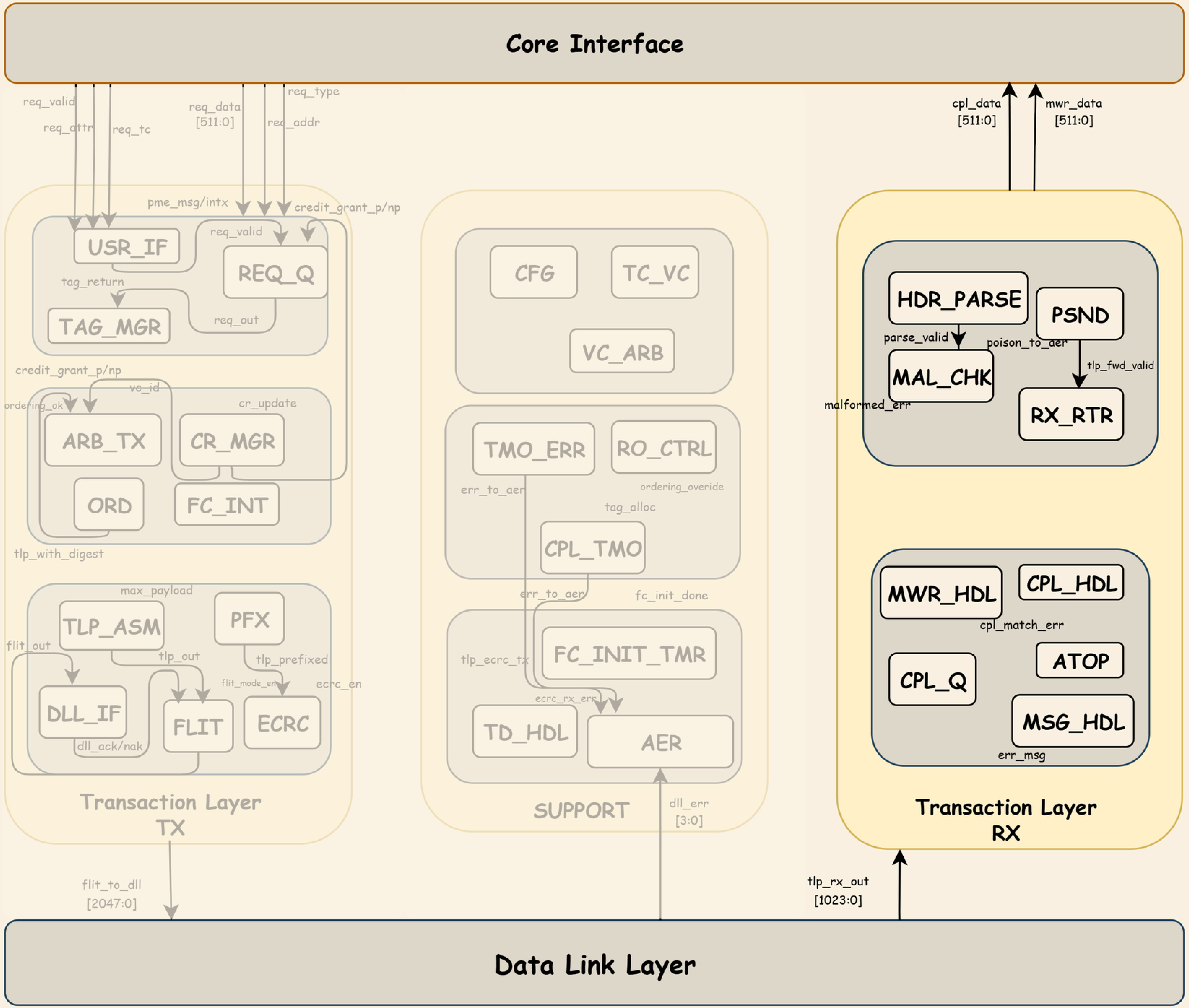
- Flow control managment .
- Virtual channels for different periorities
- Timed out completion error handling .
- Errors & Violations reporting .

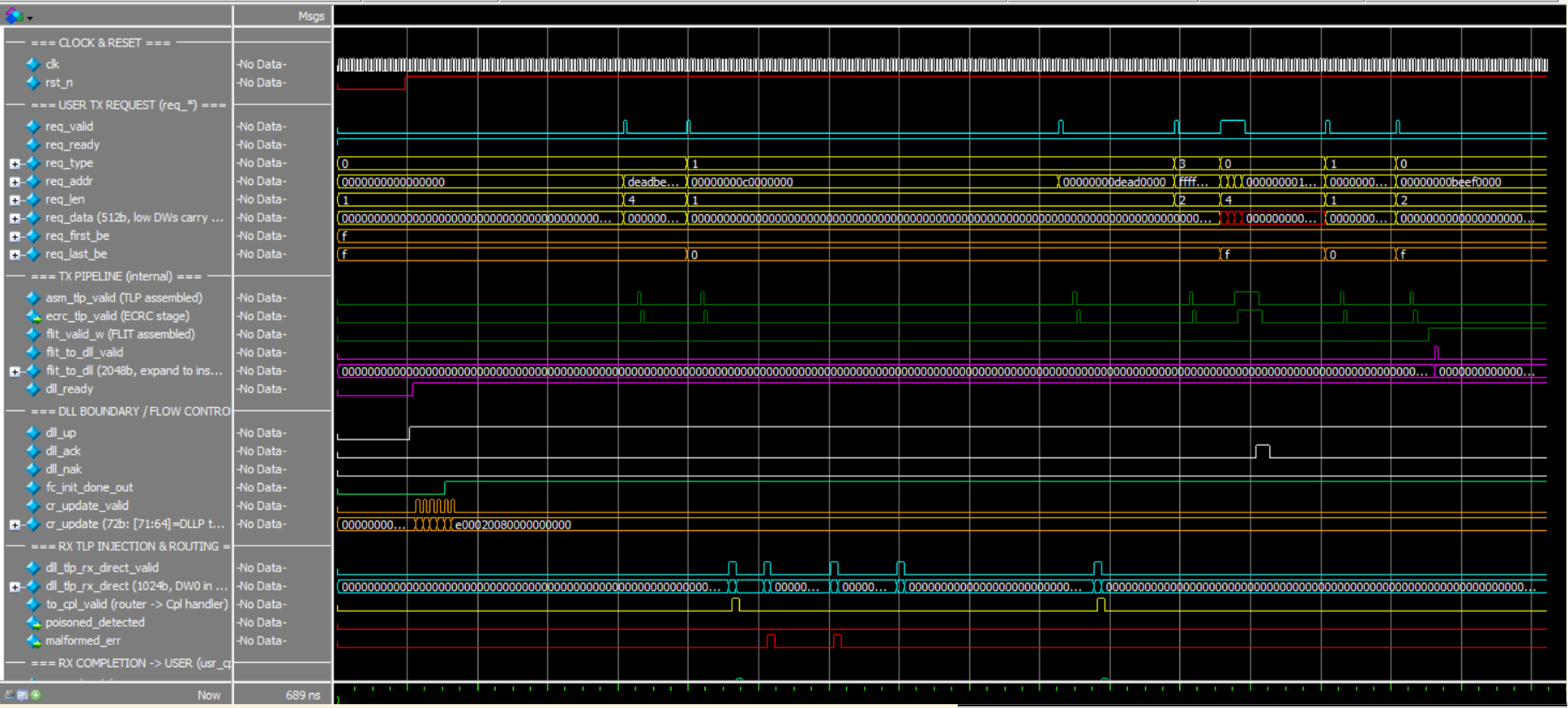


RX

ORDERING & ERROR HANDLING

- TLP Reception and Validation.
- Transaction Routing.
- Completion Reassembly and ordering
- ECRC verification.
- Processing and handling .
- Flow control updates (credits&tags)





WAVEFORM OF TRANSACTION

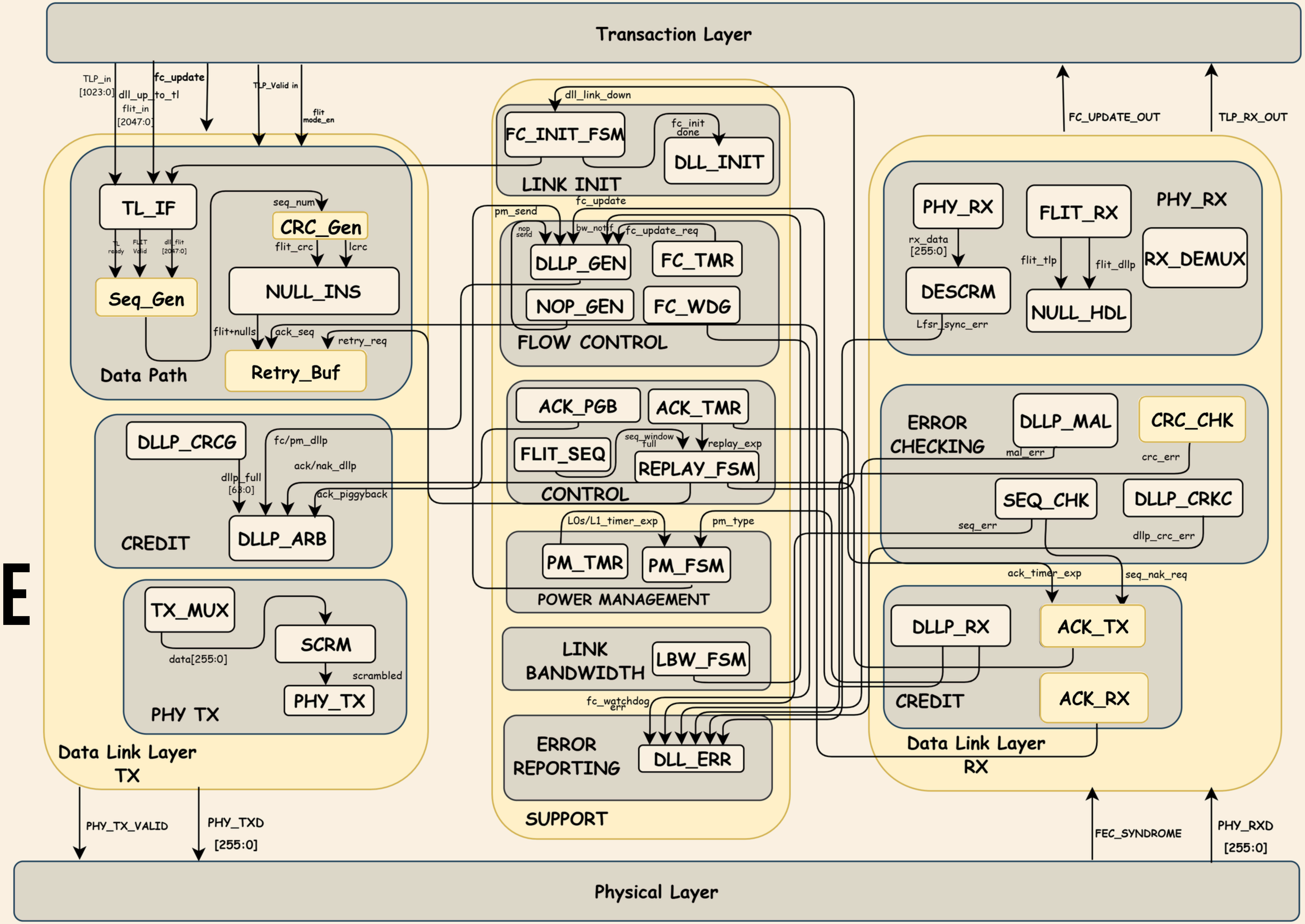
```
TEST SUMMARY
PASS: 15   FAIL: 0
==========================================
*** ALL TESTS PASSED ***
==========================================
```

TLP STRUCTURE

HEADER	DATA PAYLOAD	ECRC
12 OR 16 BYTE	0–4096 BYTES	4 BYTES

HEADER	Fmt[2:0], Type[4:0], TC[2:0], Attr, Length[9:0], Requester ID, Tag, Address
DATA PAYLOAD	Optional; 4-byte aligned DWORDs; Length field in header specifies DW count
ECRC	End-to-End CRC-32 over header+data; detects data corruption through switches

DATA LINK LAYER CORE ARCHITECTURE



TX

1
- 2-stage pipline register.
- forwards FC credits to DLLP gen.

2
- assign 12-bit seq num.
- compute CRC-24 over FLIT+SEQ.

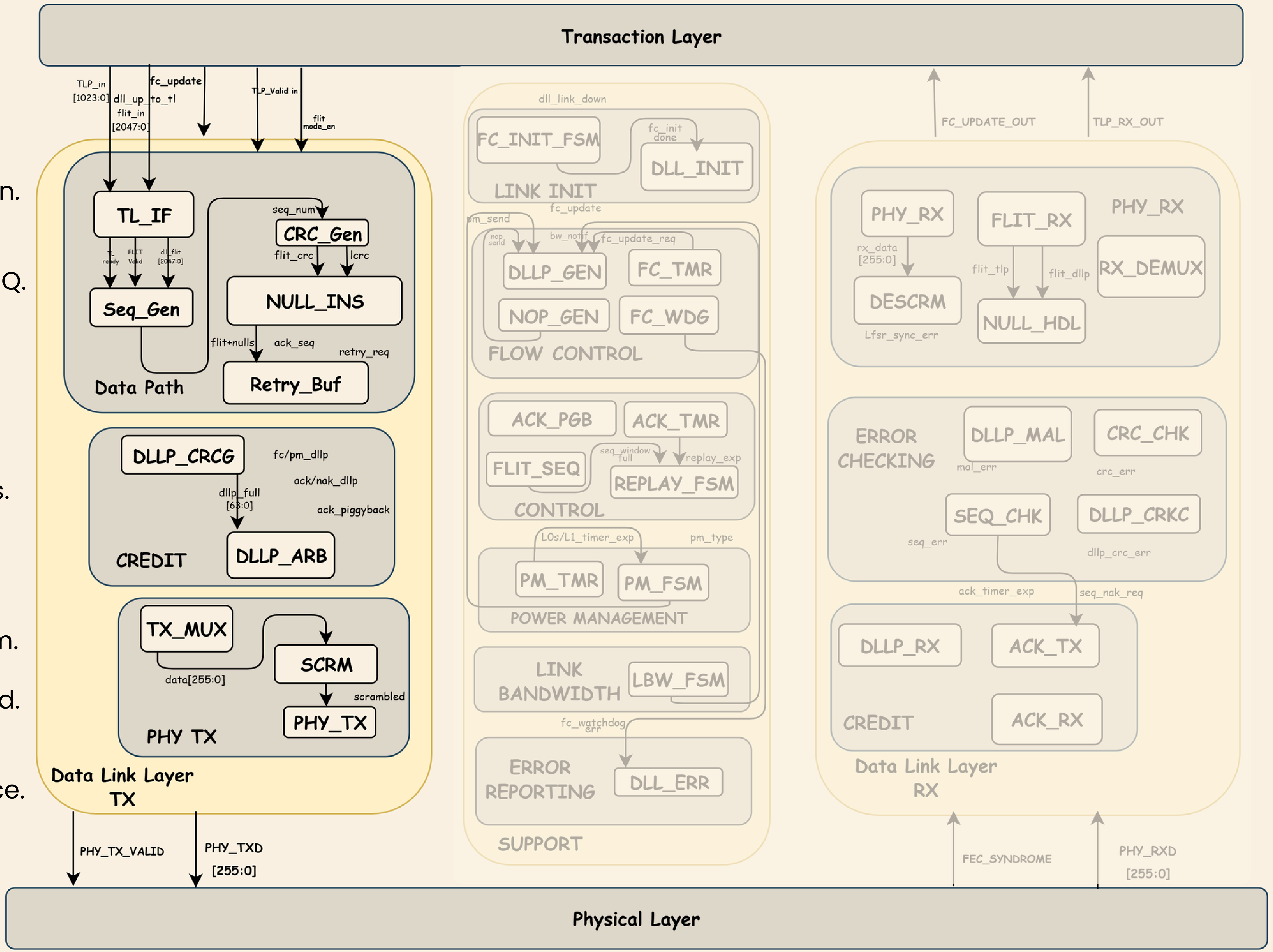
3
- keep until ACK arrives.
- Replay on NAK.

4
- priority: Retry->TLP->DLLP.
- Serialize into 256-bit beats.
- Add SOP/EOP framing markers.

5
- Replace empty 128-byte slots with null marker pattern.

6
- XOR with 23-bit LFSR keystream.
- prevents long runs of 0s or 1s.
- keep PHY clock recovery locked.

7
- 2-stage pipline to PHY.
- Handles Elec-Idle& Compliance.
- Drives phy-txd to SERDES.



SUPPORT

1 Fires dll_reset_seq pulse to reset all sequence number counters when link transitions.

2 Link cannot go Active until fc_init_done is asserted; timeout raises fc_init_err.

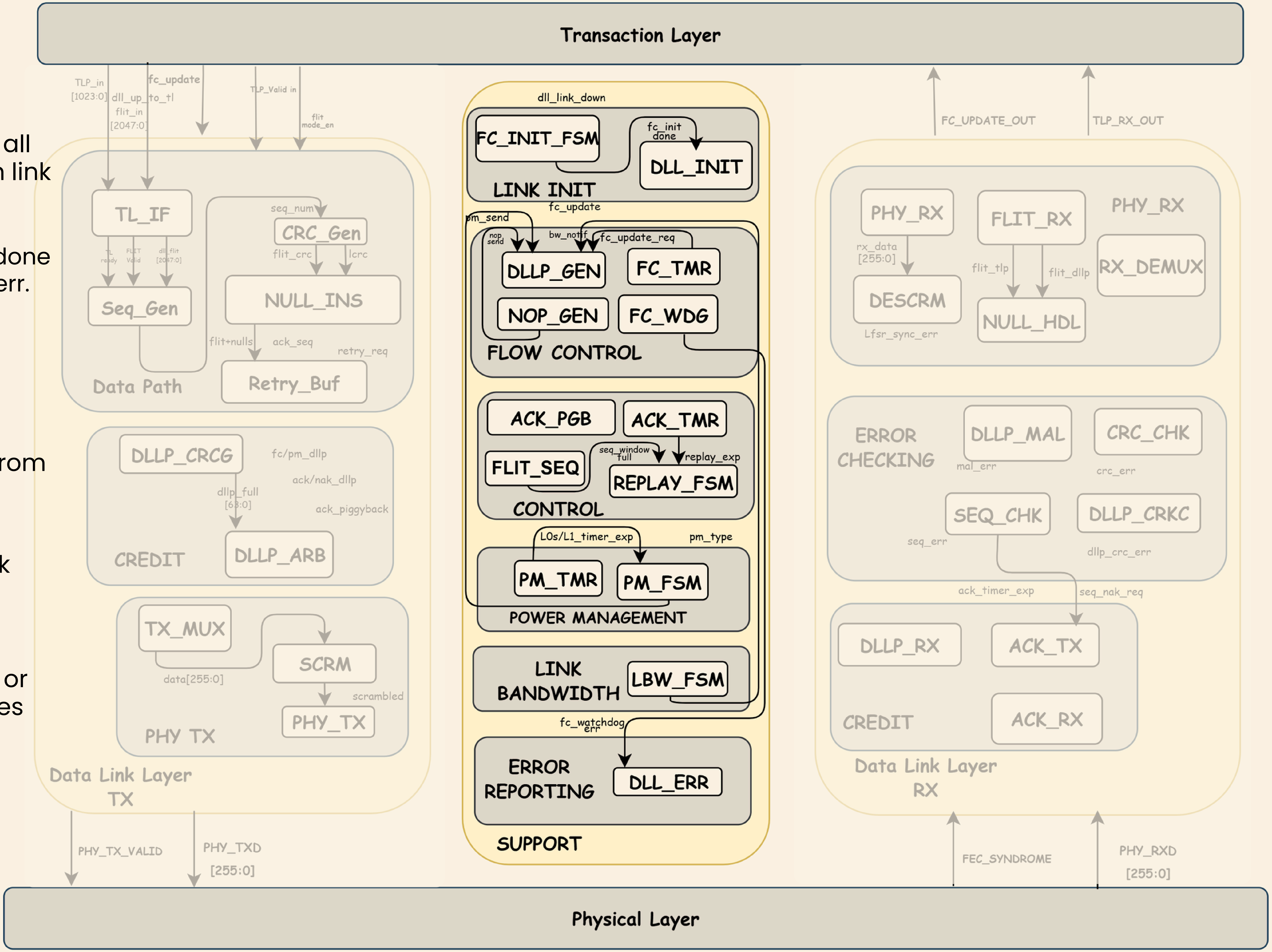
3 Handles priority when multiple requests arrive simultaneously .

4 Prevents the remote transmitter from thinking the link went silent and declaring timeout.

5 Three conditions must be true: link active, timer fired, and not suppressed by nop_inhibit

6 If no credits (Posted, Non-Posted, or Completion) arrive within limit, fires fc_deadlock_det and recovery request.

7 If NOP doesn't arrive within ACK latency limit, forces a standalone ACK; pulses ack_sent to reset timers.



RX

1. PHY reception & FLIT reconstruc
2. Descrambling & FEC validation
3. TLP / DLLP separation
4. CRC & sequence number valid
5. Flow control, ACK & NAK proces

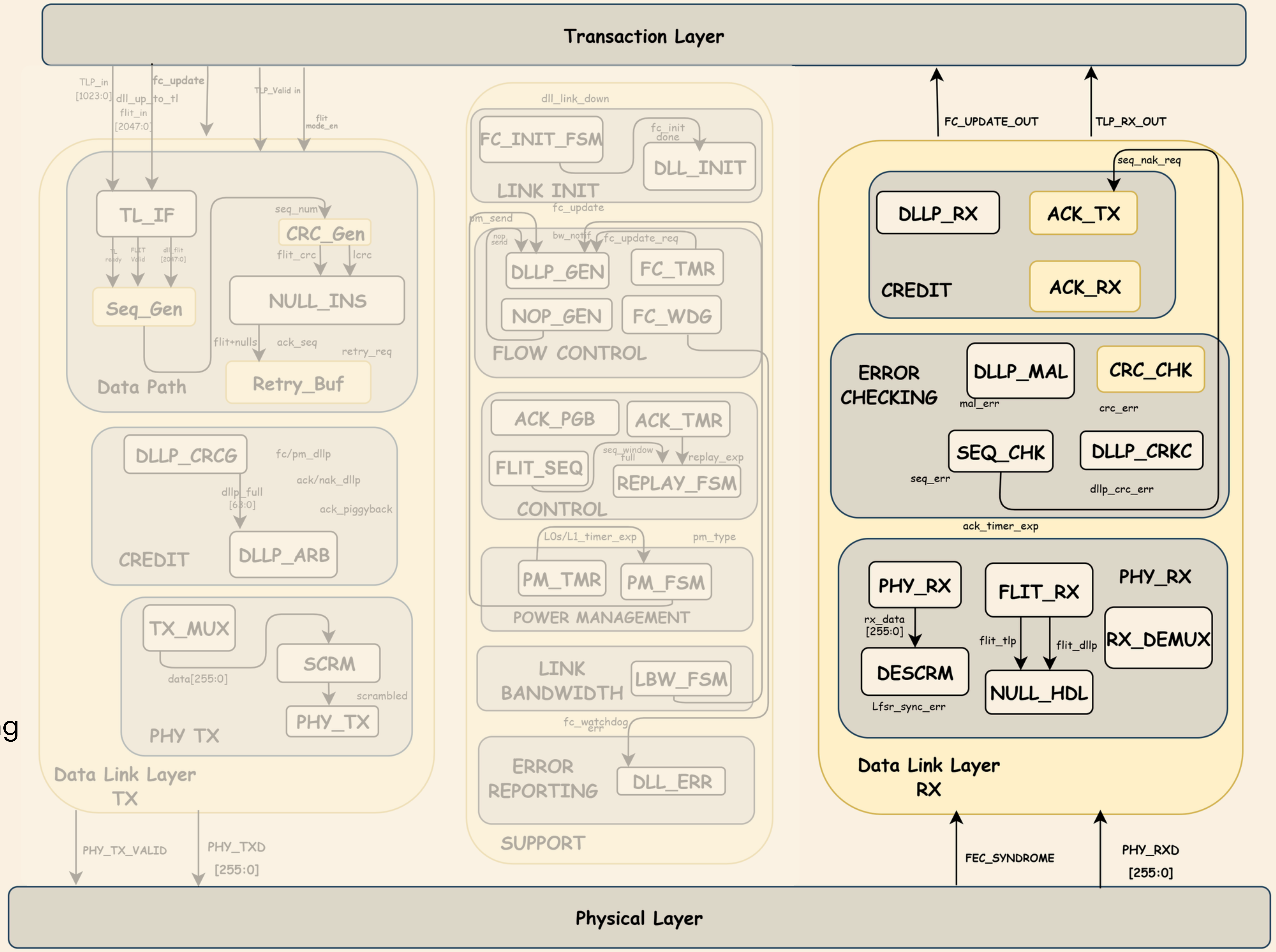
PACKET VALID?

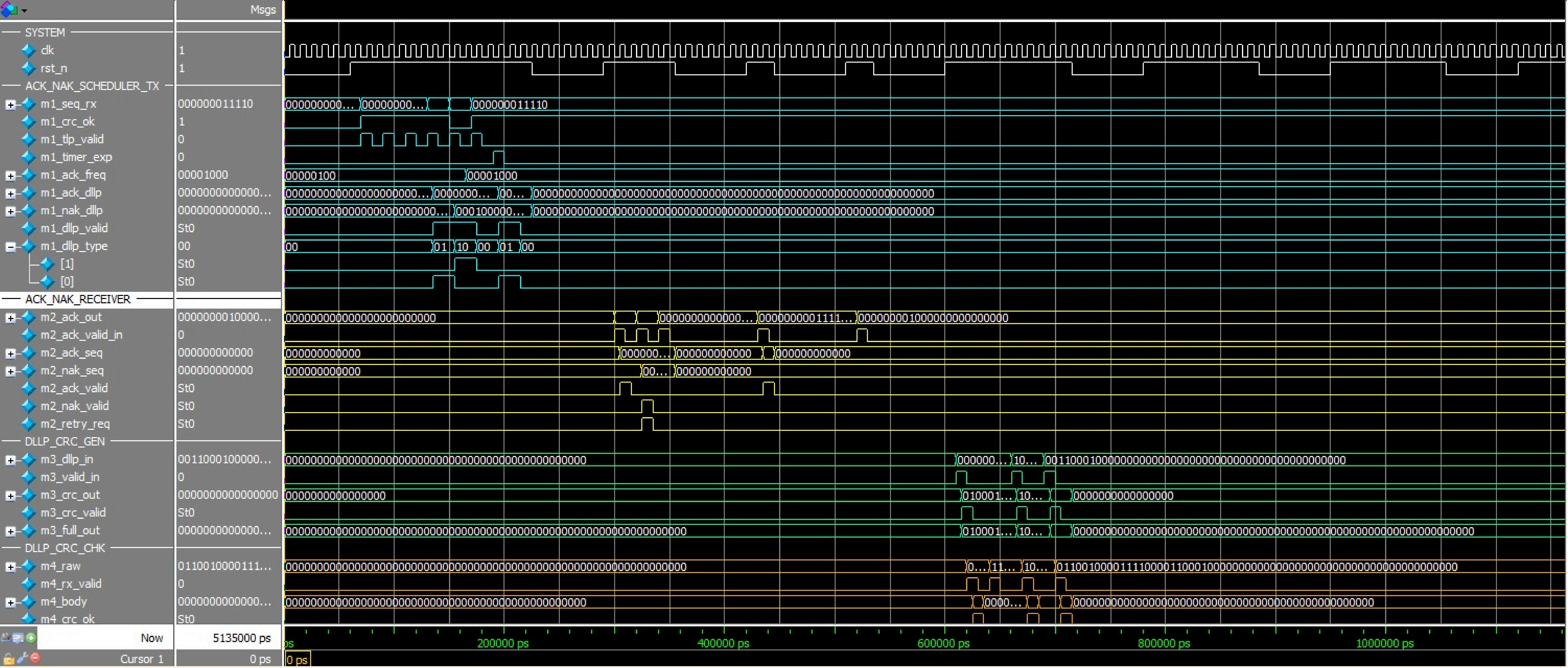
YES

Transaction Layer delivery
Valid TLPs forwarded for processing

NO

Error recovery
Triggers retrain/retry





WAVEFORM OF DLL

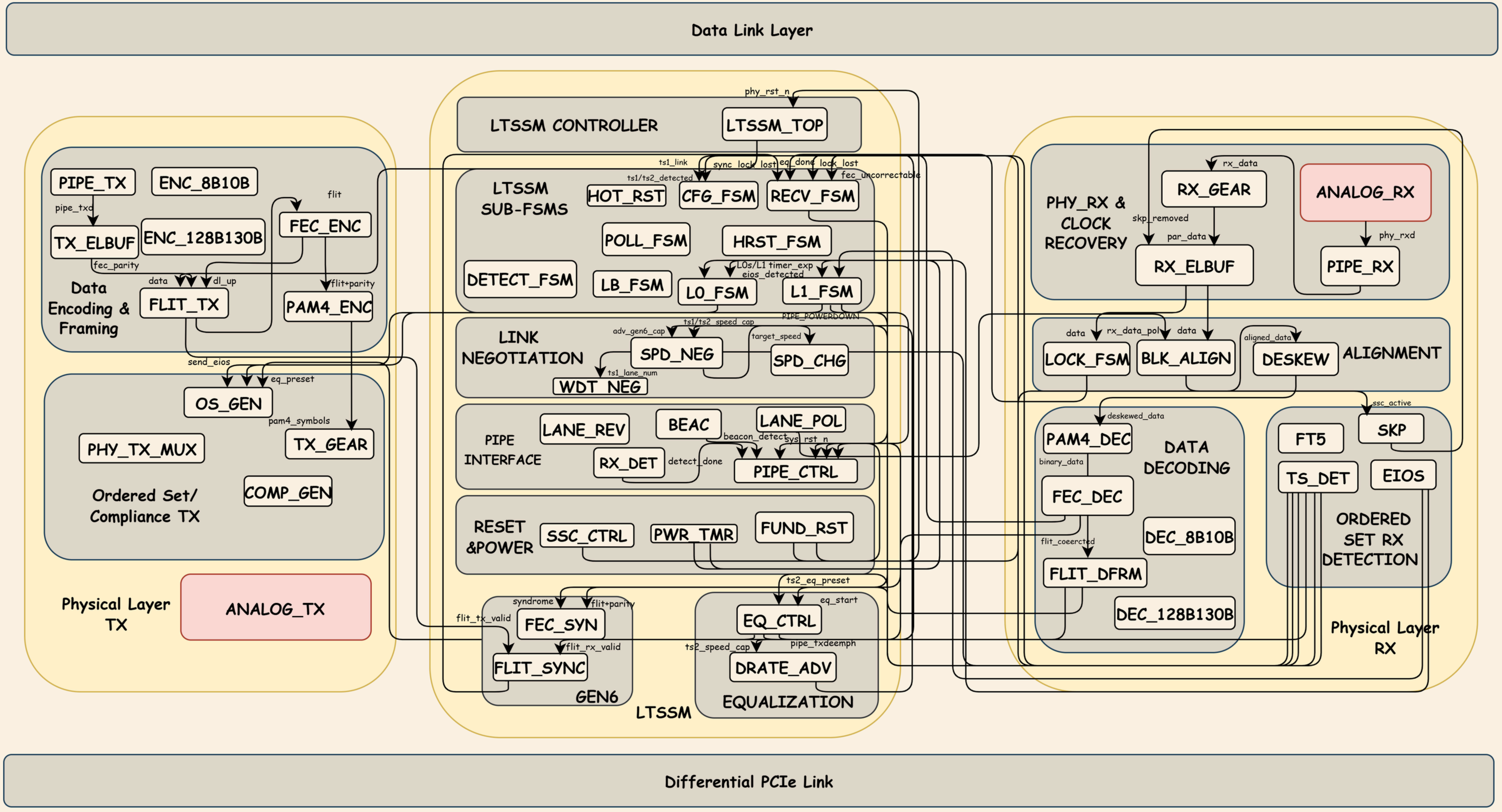
```
# ========================================
#   RESULTS: 86 PASSED  |  0 FAILED
# ========================================
# ALL 86 TESTS PASSED â
```

DLLP STRUCTURE

TYPE	PAYLOAD	CRC-16
1 BYTE	3 BYTES	2 BYTES

TYPE	DLLP type: Ack=0x00, Nak=0x10, UpdateFC-P=0x40, PM=0x20…
PAYLOAD	Type-dependent: AckSeqNum[11:0], or FC Credit HdrFC[7:0]+DataFC[11:0]
CRC-16	CRC-16 over bytes 0–3 for DLLP integrity; polynomial 0x100B

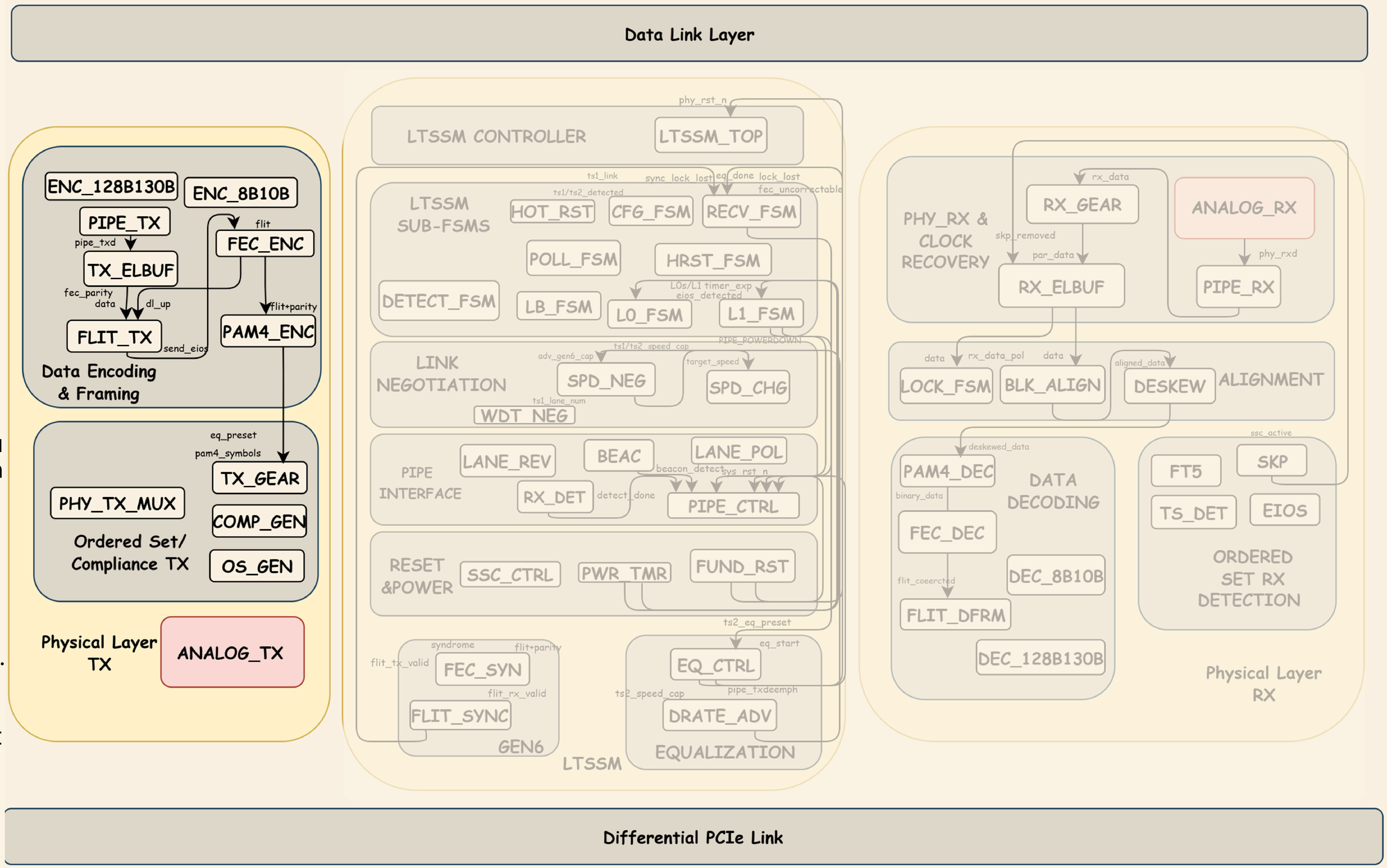
PHYSICAL LAYER CORE ARCHITECTURE



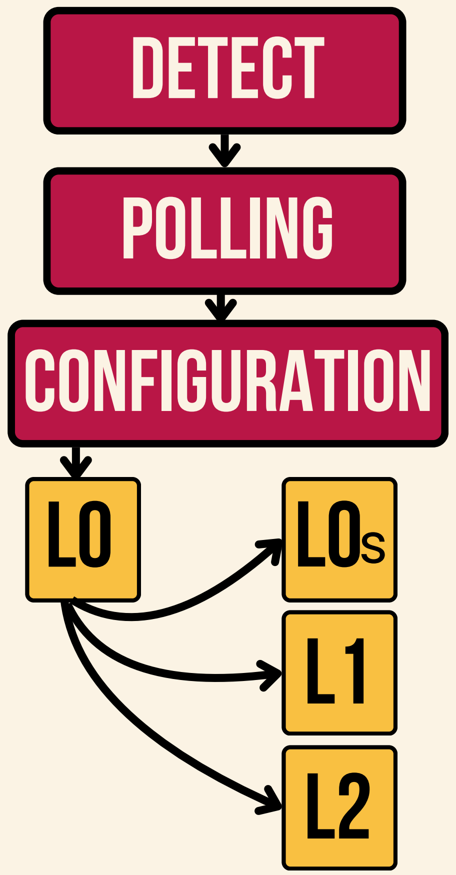
Data Link Layer
LTSSM CONTROLLER
LTSSM_TOP
phy_rst_n
LTSSM SUB-FSMS
HOT_RST
CFG_FSM
RECV_FSM
ts1_link
ts1/ts2_detected
sync_lock_lost
eq_done
lock_lost
fec_uncorrectable
POLL_FSM
HRST_FSM
DETECT_FSM
LB_FSM
L0_FSM
L1_FSM
L0s/L1 timer_exp
eios_detected
PIPE_POWERDOWN
LINK NEGOTIATION
adv_gen6_cap
ts1/ts2_speed_cap
SPD_NEG
target_speed
SPD_CHG
ts1_lane_num
WDT_NEG
PIPE INTERFACE
LANE_REV
BEAC
LANE_POL
beacon_detect
sys_rst_n
RX_DET
detect_done
PIPE_CTRL
RESET &POWER
SSC_CTRL
PWR_TMR
FUND_RST
ts2_eq_preset
eq_start
EQ_CTRL
ts2_speed_cap
pipe_txdeemph
DRATE_ADV
EQUALIZATION
syndrome
flit+parity
flit_tx_valid
FEC_SYN
flit_rx_valid
FLIT_SYNC
GEN6
LTSSM
PIPE_TX
ENC_8B10B
pipe_txd
flit
TX_ELBUF
ENC_128B130B
FEC_ENC
fec_parity
data
dl_up
flit+parity
Data Encoding & Framing
FLIT_TX
PAM4_ENC
send_eios
eq_preset
OS_GEN
pam4_symbols
PHY_TX_MUX
TX_GEAR
Ordered Set/ Compliance TX
COMP_GEN
Physical Layer TX
ANALOG_TX
PHY_RX & CLOCK RECOVERY
rx_data
RX_GEAR
ANALOG_RX
skp_removed
par_data
phy_rxd
RX_ELBUF
PIPE_RX
data
rx_data_pol
data
aligned_data
LOCK_FSM
BLK_ALIGN
DESKEW
ALIGNMENT
deskewed_data
ssc_active
PAM4_DEC
DATA DECODING
binary_data
FEC_DEC
DEC_8B10B
flit_coeercted
FLIT_DFRM
DEC_128B130B
FT5
SKP
TS_DET
EIOS
ORDERED SET RX DETECTION
Physical Layer RX
Differential PCIe Link

TX

- **PIPE_TX** — receives data from the PCIe digital logic and sends it to the TX chain.
- **Encoding (8B10B / 128B130B)** — converts data into a line code to keep the signal balanced and stable.
- **FEC_ENC** — adds extra bits for error correction at high speeds (Gen5/6).
- **PAM4_ENC** — converts the signal to 4 levels instead of 2, doubling the data rate.
- **ANALOG_TX** — the final analog driver that physically puts the signal on the PCIe lane.



LTSSM



ERROR / MAINTENANCE PATH

L0 → RECOVERY → CONFIGURATION → L0

DEBUG PATH:

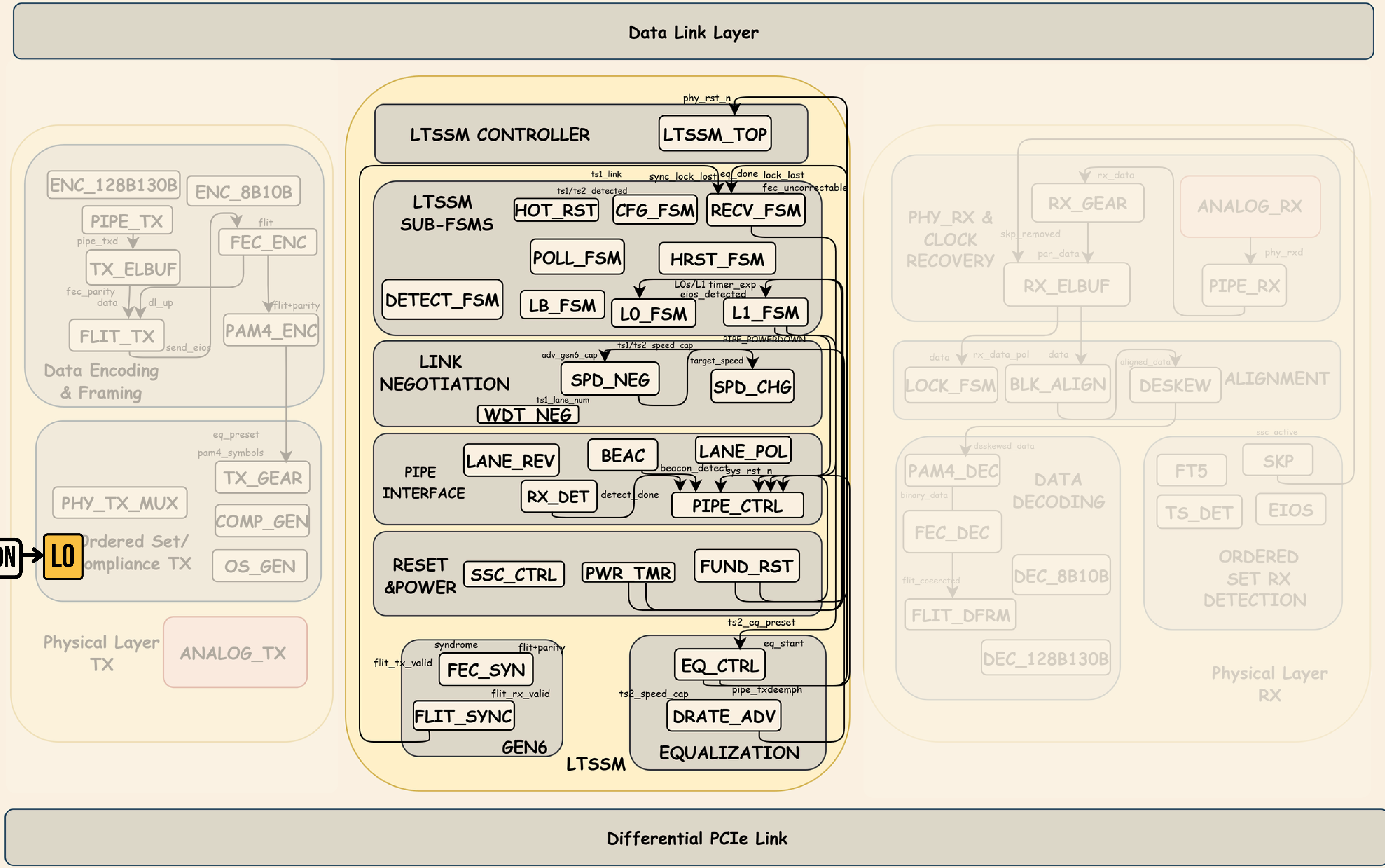
RECOVERY → LOOPBACK

DISABLE PATH:

HOT RESET → LOOPBACK

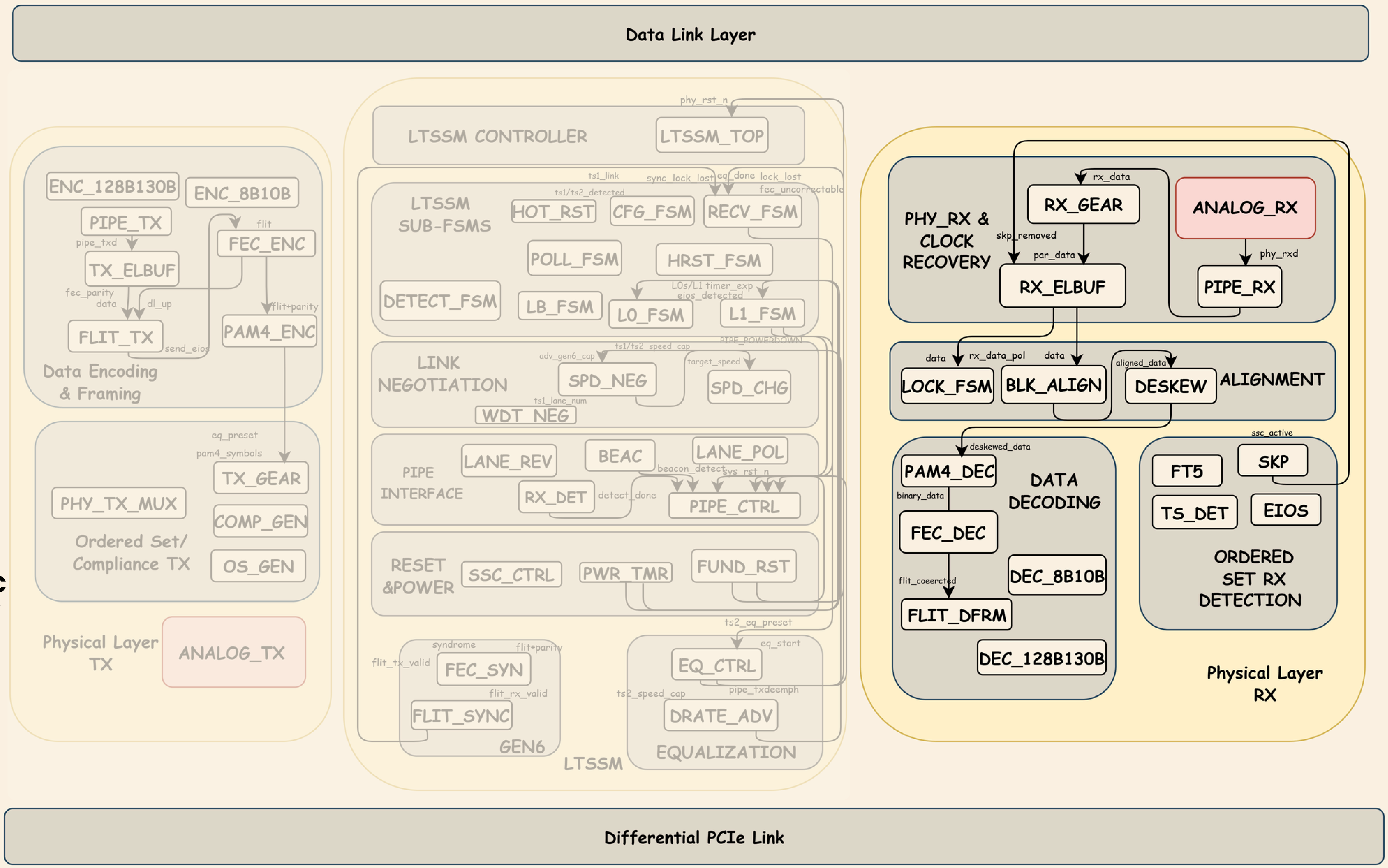
RESET PATH:

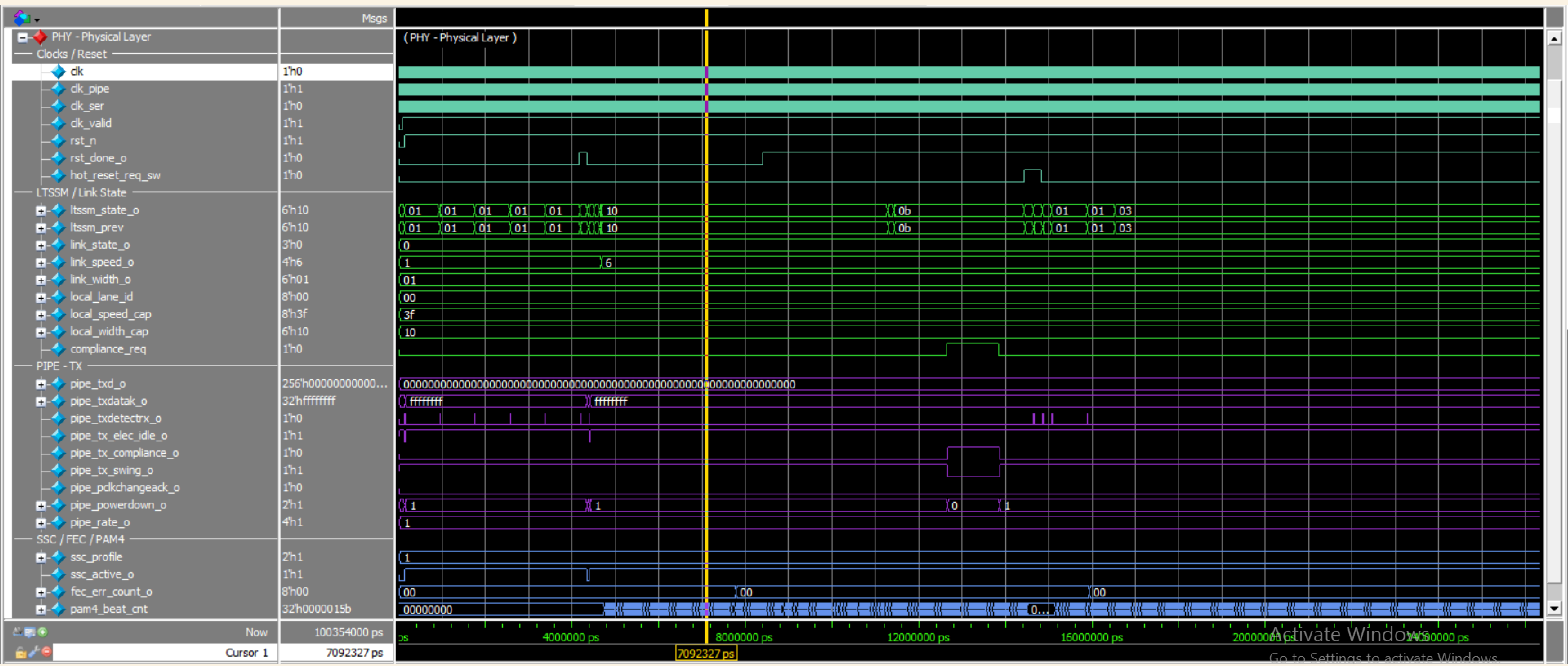
DISABLE → DETECT



RX

- **ANALOG_RX —** receives the raw analog signal coming in from the PCIe lane.
- **RX_GEAR** — adapts the data rate before it enters the digital domain.
- **DESKEW / ALIGNMENT** — fixes timing differences between lanes so all lanes arrive together.
- **PAM4_DEC + FEC_DEC** — converts PAM4 back to binary and corrects any bit errors.
- **DEC_8B10B / 128B130B** — strips the line encoding and recovers the original clean data.





WAVEFORM OF PHY

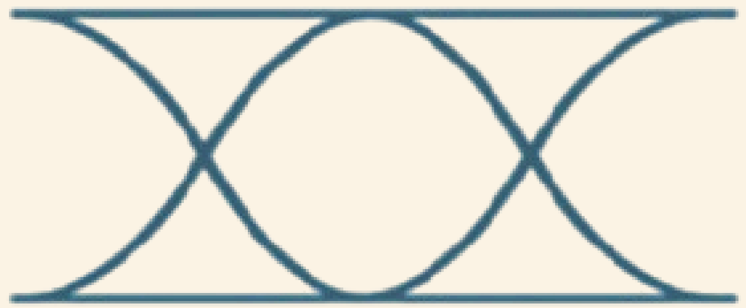
```
# ==================================================================
#   TEST REPORT
# ==================================================================
#   Total test cases : 59
#   Checks passed    : 84
#   Checks failed    : 0
#   Overall result   :        ALL PASSEDÿÿÿ
# ==================================================================
#   *** PCIe Gen6 Physical Layer â ALL TESTS PASSED ***
# ==================================================================
```

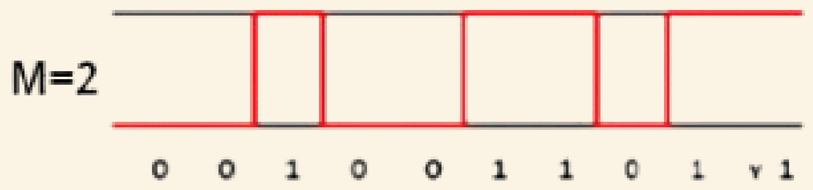
PAM4 SIGNALING CONCEPT

PAM4 — Doubling Bits Per Baud

NRZ (NON-RETURN-TO-ZERO)

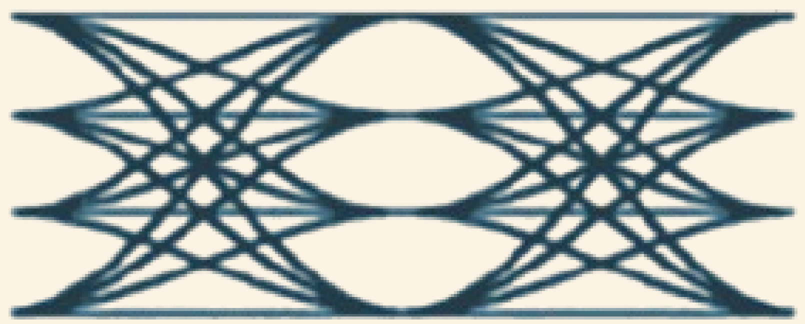
LEVELS:	2
BITS/SYMBOL:	1
EYE OPENINGS:	1
USED IN:PCIE	GEN1–5

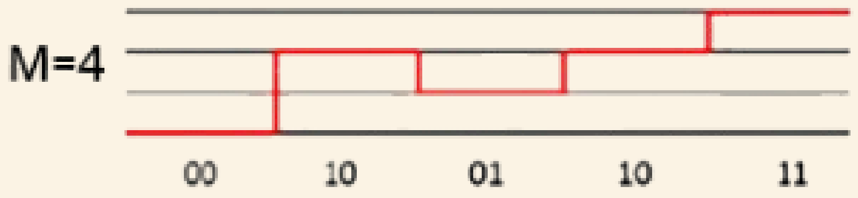




PAM4 (PULSE AMPLITUDE MOD.)

LEVELS:	4
BITS/SYMBOL:	2
EYE OPENINGS:	3
USED IN:PCIE GEN6, 400G ETHERNET	





FEC & CRC MECHANISMS

Error Control: Three Layers of Protection

- CRC (Cyclic Redundancy Check) is used to detect transmission errors in packets and FLITs.
- FEC (Forward Error Correction) enables correction of a limited number of bit errors without retransmission.
- CRC provides high reliability by identifying corrupted data.
- FEC improves link robustness, especially at high PCIe 6.0 data rates.
- Together, CRC and FEC ensure data integrity and maintain low error rates.
- These mechanisms help achieve reliable communication at 64 GT/s operation.

FLIT MODE ARCHITECTURE

FLIT (Flow Control Unit) replaces the variable-length TLP+DLLP stream with fixed containers, enabling pipelined FEC and eliminating framing overhead.

256-BYTE FLIT STRUCTURE

TLP DATA	DLP	FLIT CRC	FEC PARITY
256 BYTE	6 B	8 B	6 B

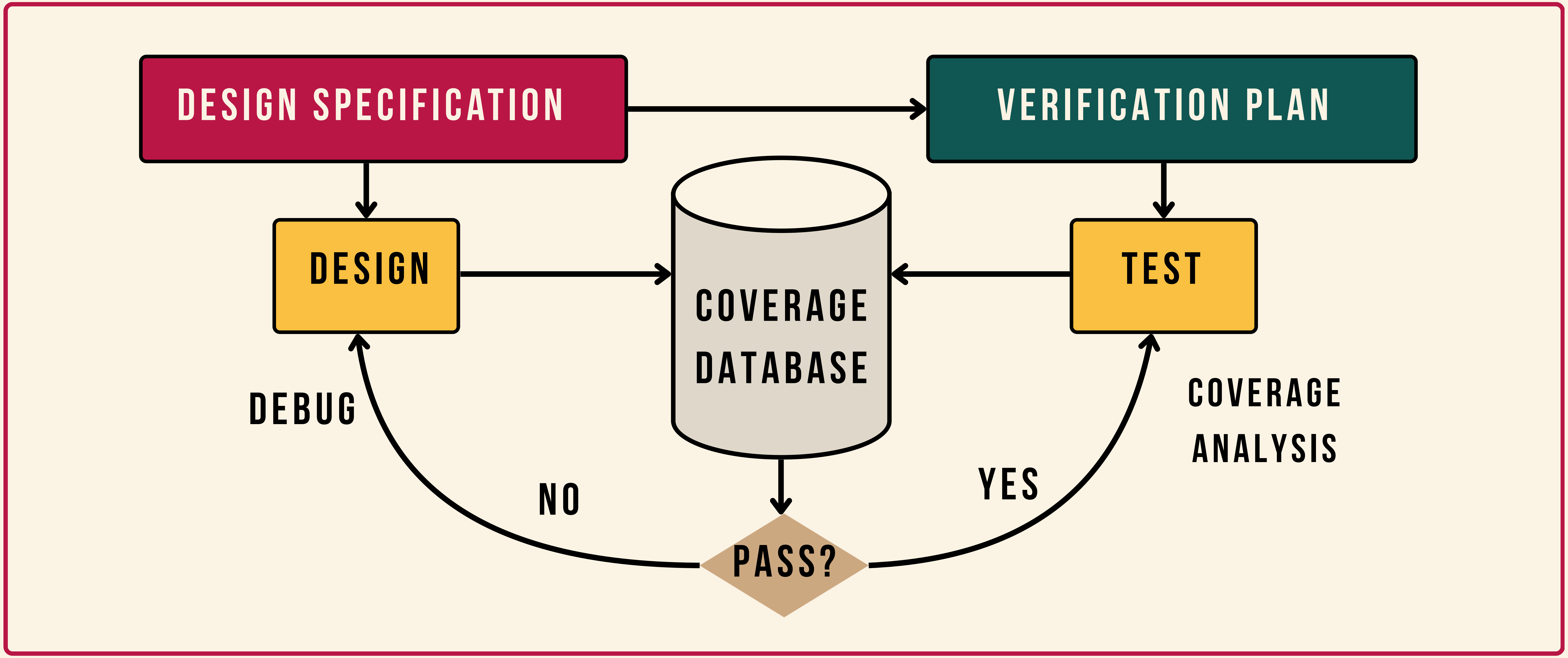
TLP	Carries TLP data and protocol information.
DLP	Contains flow control updates and data link layer information.
FLIT CRC	24-bit CRC over entire FLIT (header + payload); replaces per-TLP LCRC
FEC PARITY	Reed-Solomon parity symbols appended after FLIT CRC for RS(544,514)

COMPLETE PACKET FLOW THROUGH THE SYSTEM

TX Path

- **APPLICATION** → Issues 64-byte DMA write request
- **TRANSACTION LAYER** → Builds MWr TLP: 3DW header + 64B payload + ECRC → 80 bytes total
- **TL→DLL INTERFACE** → TLP passed via valid/ready handshake; FC check: Posted credit consumed
- **DATA LINK LAYER** → Prepends 4B DLL wrapper (Seq#) → stores in retry buffer → appends FLIT CRC
- **DLL→PHY INTERFACE** → 256B FLIT assembled; FC delta encoded in FLIT header
- **PHYSICAL LAYER** → Scramble → 128b/130b encode → RS-FEC parity append → PAM4 symbol map
- **SERDES (PIPE)** → 32 GBaud PAM4 symbols transmitted; MSB/LSB lanes interleaved

VERIFICATION AND COVERAGE ANALYSIS WORKFLOW



VERIFICATION TEST PLAN

UNIT-LEVEL VERIFICATION → BLOCK-LEVEL VERIFICATION → SYSTEM-LEVEL VERIFICATION

TRANSACTION LAYER TESTS

- All TLP types coverage
- FC credit stall and recovery
- Max payload (4096B) MWr/MRd
- 1024 simultaneous outstanding tags
- Poisoned TLP injection and upstream reporting

DATA LINK LAYER TESTS

- ACK/NAK basic handshake
- Retry buffer fill and drain
- Sequence number wrap-around
- NAK storm (10 consecutive NAKs)
- Replay timer expiry and retransmission

PHYSICAL LAYER TESTS

- Full LTSSM traversal: Detect→L0
- Link width negotiation (x1/x4/x8/x16)
- Scrambler sync and reset
- RS-FEC single/multi-symbol error correction
- L0→L0p→L0 low-power transition

INTEGRATION & STRSS TESTS

- Simultaneous bidirectional traffic (x16)
- Mixed TLP types with error injection
- Link recovery during active transfer
- 10M-packet stress run (no data corruption)

VERIFICATION METHODOLOGY

WE CHOSE A LAYERED, COMPONENT-BASED VERIFICATION APPROACH USING NATIVE SYSTEMVERILOG , ENABLING FASTER SIMULATION COMPILE TIMES AND DIRECT DESIGN INSIGHT.

- DIRECTED TESTS
- ERROR INJECTORS
- ASSERTIONS (SVA)
- CONSTRAINED RANDOM
- MONITORS & SCOREBOARDS
- COVERAGE GROUPS

```
===================================================================
  PCIe Gen6 SystemVerilog Testbench v1.0  â FINAL SUMMARY
===================================================================
  Test Cases: 55   |   Checks PASSED: 88     |   FAILED: 0
===================================================================
  RESULT:  ALL CHECKS PASSED  â System verified
===================================================================

[COVERAGE] LTSSM coverage: 100.0%
[COVERAGE] AER   coverage: 100.0%
[COVERAGE] TLP   coverage: 100.0%
===================================================================
```

VERIFICATION ENVIRONMENT ARCHITECTURE

GENERATE STIMULUS

Create directed, random, and constrained-random test scenarios

APPLY STIMULUS TO THE DUT

Send generated stimulus to the PCIe design

GOLDEN MODEL CREATION

to generate expected outputs from design in TE

CAPTURE THE RESPONSE

Monitor and collect DUT outputs and protocol activity

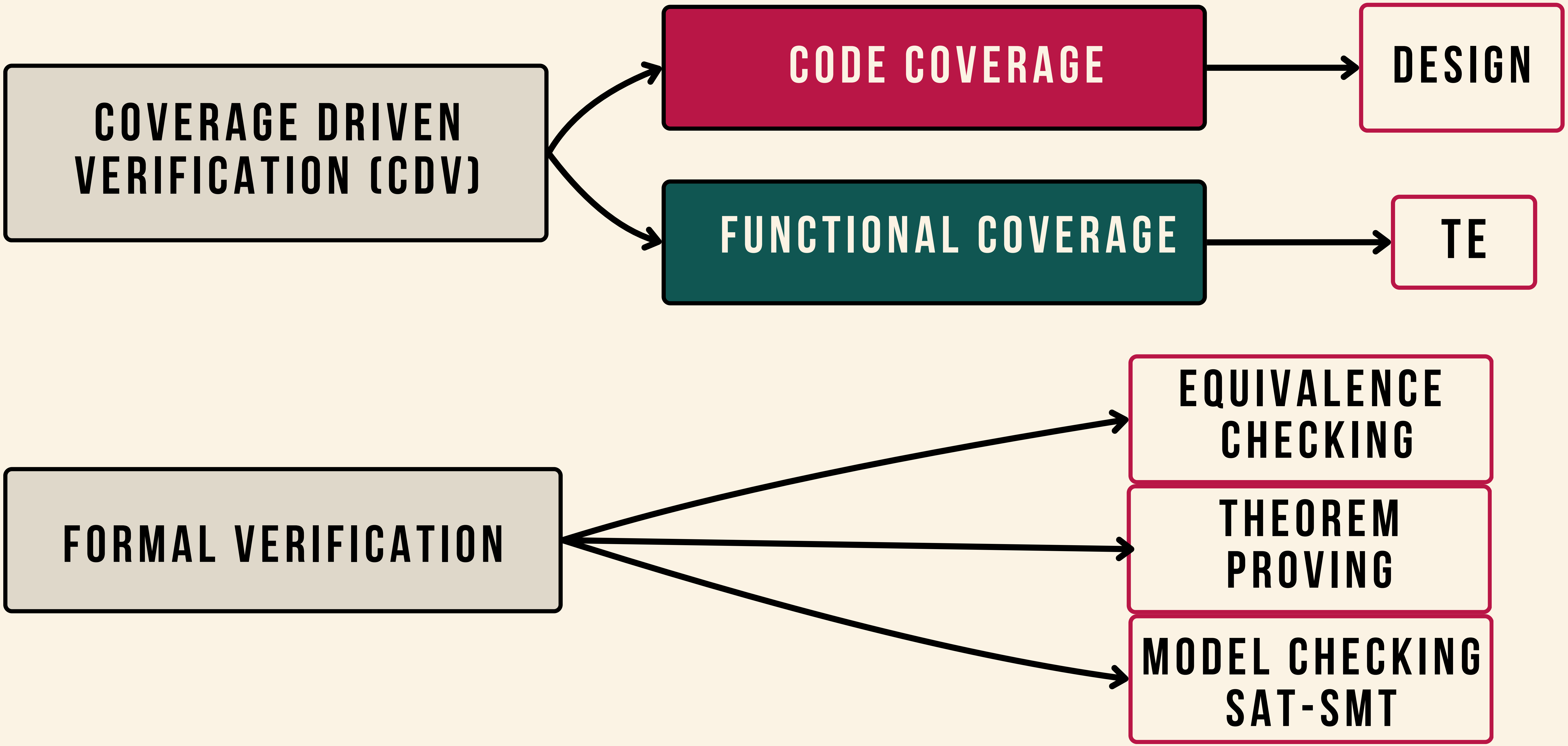
CHEACK FOR CORRECTNESS

Compare actual results with expected results using assertions and scoreboards

MEASURE PROGRESS AGAINST THE DUT (3-LAYER PCIE)

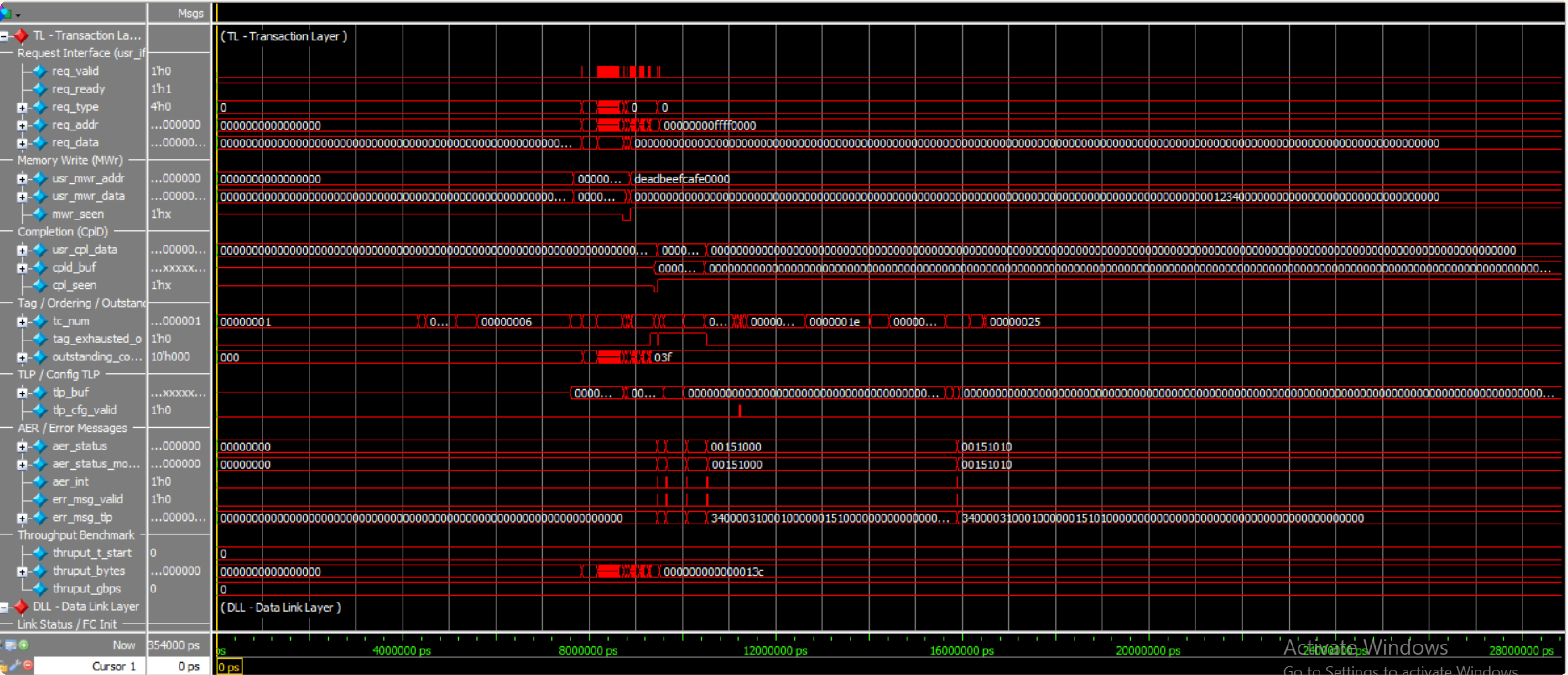
Design Under Test: TL +DLL + PHY-D RTL

COVERAGE-DRIVEN VERIFICATION METHODOLOGY

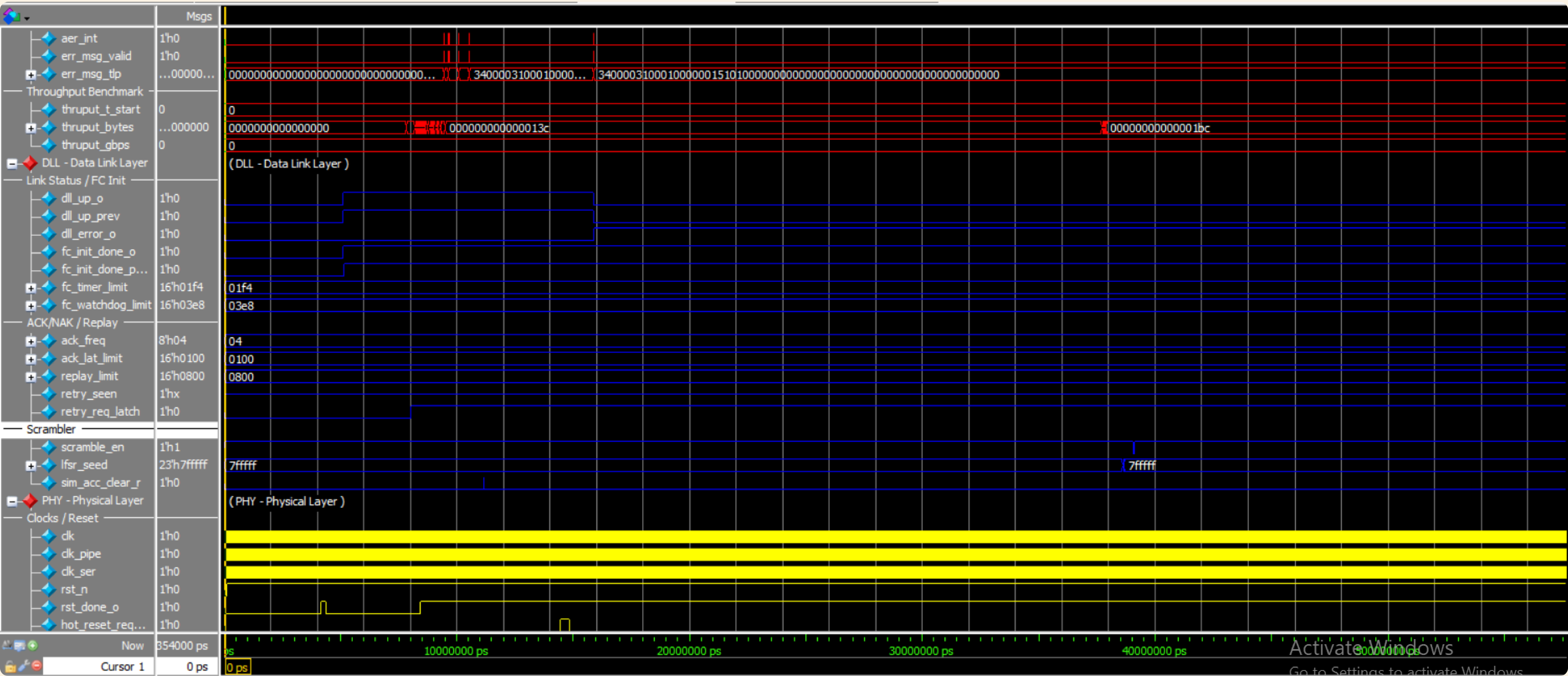


VERIFICATION RESULTS

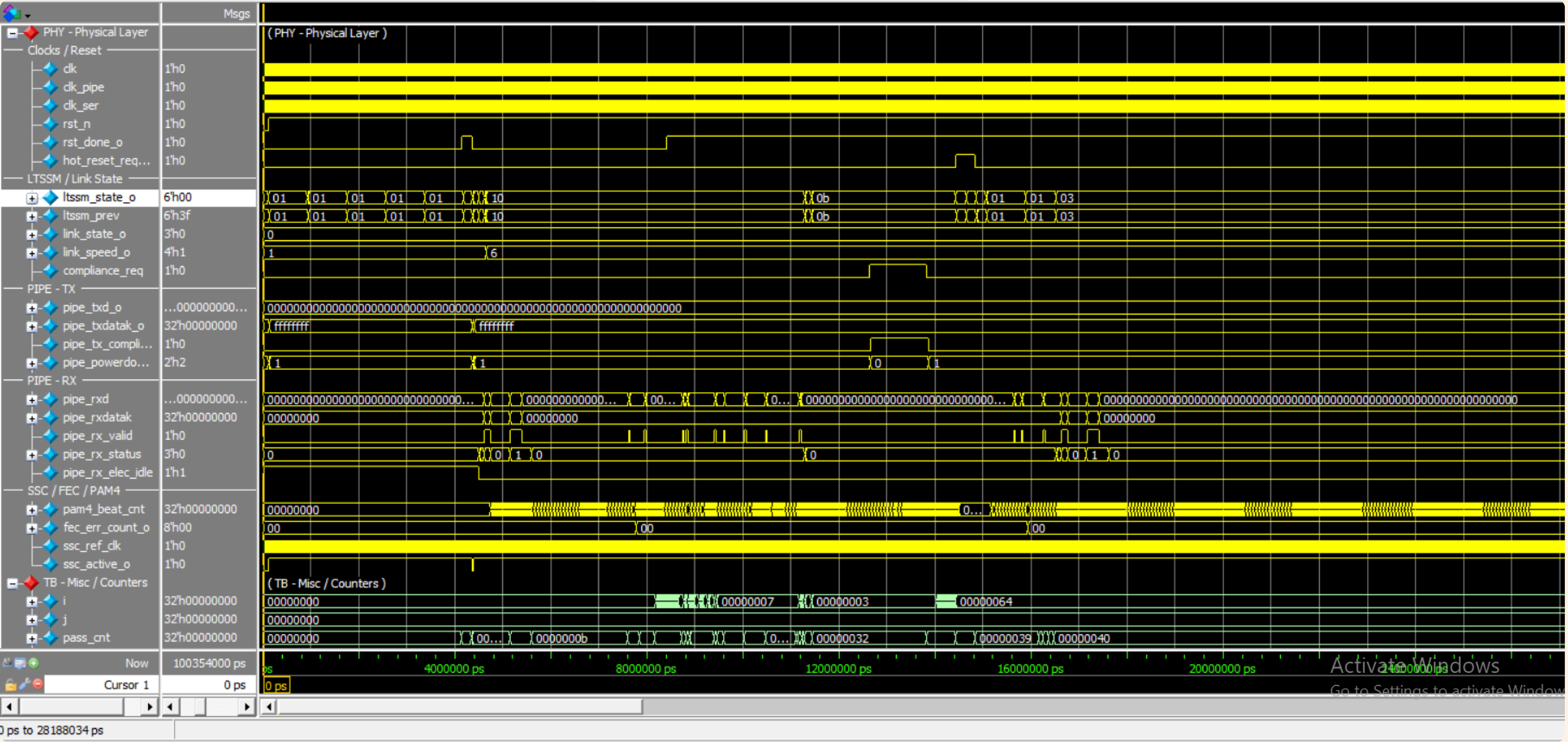
TL-TRANSACTION LAYER



DLL-DATA LINK LAYER



PHY-PHYSICAL LAYER



QUESTA COVERAGE REPORT

Link Width	Aggregate Bytes	Window (ns)	Achieved (Gb/s)	Theoretical Peak (Gb/s)	Efficiency (%)
×1	3,200	480	53.33	60.5	88.1
×2	6,400	480	106.67	121.0	88.2
×4	12,800	480	213.33	242.0	88.2
×8	25,600	480	426.67	484.0	88.2
×16	51,200	480	853.33	968.0	88.2

Total Coverage (100%)

Overall Design Unit Coverage Summary:

Covergroups
100%

Coverage Summary by Design Unit:

Design Unit	Covergroups	Total
work.tb_pcie_gen6_sv	100%	100%

Covergroups	Bins	Hits	Misses	Goal	Coverage
/tb_pcie_gen6_sv/cg_ltssm	9	9	0	100	100%
\/tb_pcie_gen6_sv/cov_ltssm	9	9	0	100	100%
/tb_pcie_gen6_sv/cg_aer	8	8	0	100	100%
\/tb_pcie_gen6_sv/cov_aer	8	8	0	100	100%
/tb_pcie_gen6_sv/cg_tlp_type	3	3	0	100	100%
\/tb_pcie_gen6_sv/cov_tlp	3	3	0	100	100%

PROJECT ACHIEVEMENTS

- **COMPLETE 3-LAYER PCIE 6.0 RTL**

  ~8,500 lines of RTL implementing all three layers per PCIe 6.0 Base Spec r1.0

- **NOVEL FLIT-FEC INTEGRATION**

  First open-reference implementation of FLIT mode with RS(544,514) boundary alignment

- **100% FUNCTIONAL COVERAGE**

  Exceeds industry baseline of 90% for tape-out sign-off; 23 covergroups, 175 bins

- **100% ASSERTION PASS RATE**

  All 84 SVA assertions pass across 2.4 billion simulation cycles

- **SYNTHESIS-READY RTL**

  Design synthesizes cleanly at 250 MHz in 28nm CMOS (Synopsys DC estimate)

- **100% FORMAL COVERAGE FOR TL & DLL**

  Mathematically proving the absolute functional correctness of these critical modules.

- **COMPREHENSIVE DOCUMENTATION**

  Micro-architecture spec, verification plan, and coverage closure report — 180 pages total

CONCLUSION

THIS PROJECT WAS MORE THAN AN ACADEMIC REQUIREMENT—IT WAS AN OPPORTUNITY TO APPLY ENGINEERING CONCEPTS, STRENGTHEN OUR TECHNICAL SKILLS, AND GAIN HANDS-ON EXPERIENCE WITH ADVANCED COMMUNICATION SYSTEMS. THANK YOU FOR YOUR TIME AND ATTENTION.

Q & A

THANK YOU